

CPSC 544

Advanced Software Process

Part II: The Repeatable Process

5. Managing Software Organizations

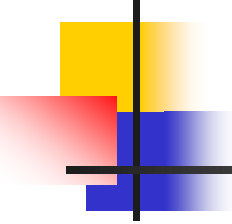


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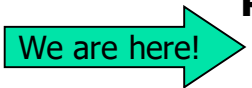


Outlines

Part I: Software Process Maturity

1. A Software Maturity Framework
2. The Principles of Software Process Change
3. Software Process Assessment
4. The Initial Process

Part II: The Repeatable Process



5. Managing Software Organizations

6. The Project Plan
7. Software Configuration Management (Part I)
8. Software Quality Assurance

Part III: Defined Process

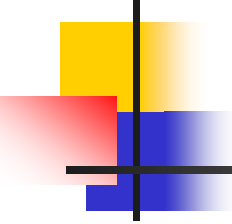
9. Software Standards
10. Software Inspections
11. Software Testing
12. Software Configuration Management (continued)
13. Defining the Software Process
14. The Software Engineering Process Group

Part IV: The Managed Process

15. Data Gathering and Analysis
16. Managing Software Quality

Part V: The Optimizing Process

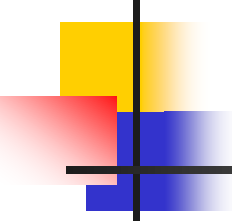
17. Defect Prevention
18. Automating the Software Process
19. Contracting for Software
20. Conclusion



Part II Summary

The Repeatable Process

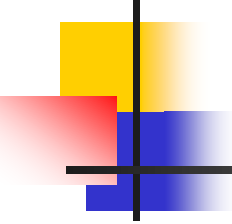
- Ch.05 – **Managing SW Organizations**
 - Deals with the way SW commitments are made and managed
- Ch.06 – **The Project Plan**
 - Reviews SW planning principles and goals, the Work Breakdown Structure, size estimating, resource estimating, and project tracking
- Ch.07 – **SW Configuration Management (Part I)**
 - Explains why configuration management is important and describes the capabilities with highest initial priority.
- Ch.08 – **SW Quality Assurance**
 - Describes the benefits and goals of SQA, how it is organized, and some of the key considerations in establishing and managing an SQA organization.



Part II Summary

The Repeatable Process

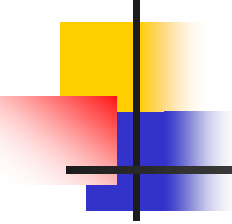
- The reason the four topics of Part II are addressed at this point is not that they cover all important issues but that they likely represent the highest-priority areas for organizational improvement at this point.



Part II Summary

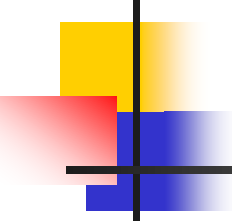
The Repeatable Process

- The reason these four topics are selected for attention at this point is that they establish a sound basis for project planning and process management.
 - Until the organization effectively handles commitments, planning, configuration management, and Quality Assurance, the process will be **too erratic to permit orderly improvement**.



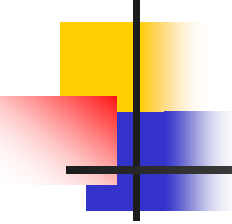
Managing SW Organizations

- Commitment Discipline
- The Management System
- Establishing a Project Management System



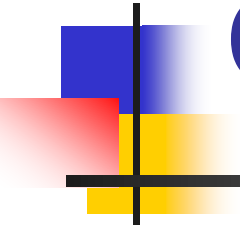
Managing SW Organizations

- The role of **the management system** is to ensure that projects are successfully completed.
- This implies some **organization-wide agreement on the meaning** of the terms “**success**” and “**completion**”.
- It also requires **a continuing management focus** on the progress of each project.
- **Project management** starts with **a definition of the job** to be done and **the plan** to do it.

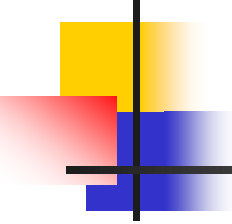


Managing SW Organizations

- Basic **principles** of **project management**:
 - Each project has a **plan** that is based on a hierarchy of commitments.
 - A **commitment** is an agreement by one person to do something for another.
 - A **management system** resolves the natural conflicts between the projects and between the line and staff organizations.
 - An **oversight and review system** audits and tracks progress against the plans.

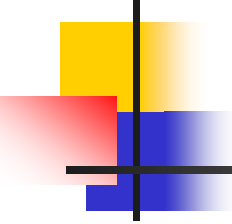


Commitment Discipline



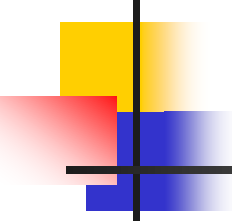
Commitment Discipline

- The foundation for SW project management is the commitment discipline.
- This is supported by plans, estimates, reviews, and tracking systems, which focus on ensuring that the organization meets its commitments.
- Commitments are not met by reviews, procedures, or tools, however; they are met by committed people.



Making a Commitment

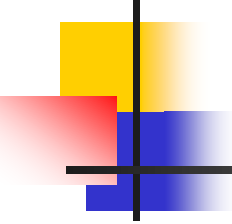
- A commitment involves a planned completion date and some consideration or payment.
- When the coordinated efforts of many professionals are involved, mutual commitments are essential.



Making a Commitment

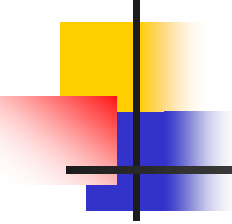
The elements of an effective commitment

- The person making the commitment does so **willingly**.
- The commitment is **not** made **lightly**.
 - The work involved, the resources, and the schedule are carefully considered.
- There is **agreement** between the parties on **what is to be done, by whom, and when**.
- The commitment is **openly and publicly stated**.
- The person **responsible** tries to meet the commitment, even if **help** is needed.
- Prior to the committed date, if it is clear that it cannot be met, **advance notice** is given and a **new commitment** is negotiated.



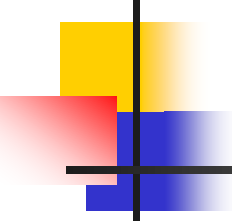
The Commitment Hierarchy

- While commitments are met by committed individuals, the work is not done in a vacuum.
- As long as the professionals feel they are part of an organization that treats its commitments seriously, they will strive to do their part.
- This calls for a management team that takes care in making commitments and then insists on extraordinary efforts to meet them.



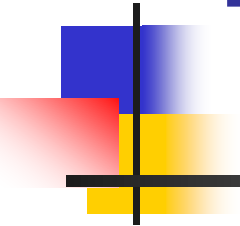
The SW Commitment Process

- To be effective, the SW commitment process must reach to the top of the organization.
- This requires that:
 - All commitments for future SW delivery are personally made by the organization's senior executive.
 - These commitments are made only after successful completion of a formal review and concurrence process.
 - An enforcement mechanism ensures that these reviews and concurrences are properly conducted.

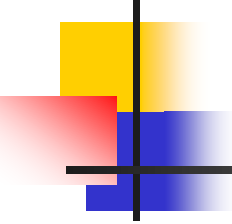


Establishing a Commitment Process

- A commitment process can be established very quickly.
- The basic requirement is a senior executive who is willing to insist that the required planning be done before any commitment is made.
- In addition, since the people must know how to make schedules and estimates, training courses are required, as are specific estimating, review, and approval procedures.

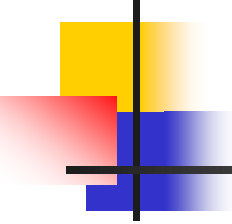


The Management System



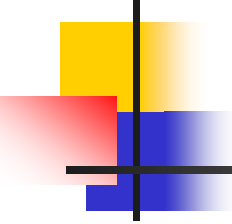
The Management System

- While every organization will have unique **objectives**, there are generally **four** components:
 - To have a technical and business strategy that aims at such long-term goals as growth rate or market position
 - To provide quality products that meet customer's needs in a timely and effective way
 - To perform the assigned mission competitively and economically
 - To improve continually the organization's ability to handle more challenging work



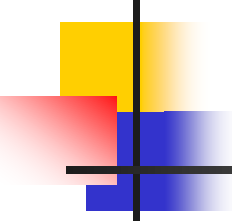
Product and Period Plans

- To resolve the inherent conflict and to establish a framework for operations, most organizations produce **annual operating plans**.
- These specify the tasks to be performed and assign the responsibilities and resources to accomplish them.
- Since **skilled resources** are the single most important need of every project and staff, their **allocation** is the essential first step in producing these plans.



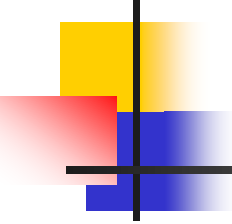
Management Oversight

- An effective management system typically uses **reviews and a contention system** to resolve product and period conflicts and establish the balance between line and staff.
- Each line and staff organization prepares its **annual plan** and **reviews** it with all involved parties.



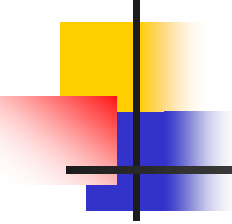
Management Oversight

- The issues are then resolved, **the separate plans** are consolidated into a total organization plan, and this total plan is incorporated in the plan for the next higher organizational level.
- Similarly, each **project area** establishes its own plan prior to project initiation and periodically reviews and updates it.



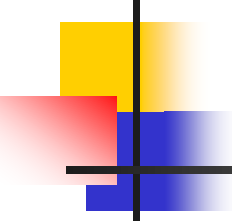
The Contention Process

- An effective **review system** requires a parallel contention system to encourage the open expression of differences and their rational resolution.
- The principle behind **the contention system** is that the best decisions are based on a full understanding of the relevant issues.
- There are many ways to solve most problems and there is rarely agreement on the best approach.



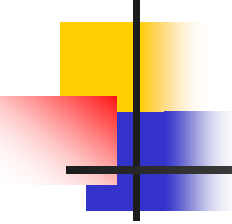
The Contention Process

- Most of successful organizations have found that the contention system helps them arrive at better decisions.
- Alfred P. Sloan, the founder of GM
 - “No important decision should be made unless there is some contention.”



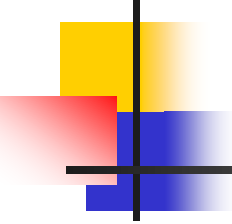
The Principles of the Contention System

- All major decisions are reviewed with the involved parties in advance, and the parties are requested to agree.
 - When possible, any issues are resolved before proceeding.
- When the time comes for the decision, all dissenting parties are present and asked to state their views.
- When there is no disagreement, the senior manager determines if there is knowledgeable agreement, if any disagreeing parties are absent, or if more preparation is needed.
 - In the latter two cases, the decision is deferred until the necessary homework has been done.



The Quarterly Review

- The quarterly review provides a forum for resolving conflicts and monitoring progress against period and product objectives.
- The topics should typically include an assessment of project performance against plan and the organization's performance against its goals.



The Quarterly Review

- Table 5.1
Example of
a Quarterly
Review
Agenda

TABLE 5.1
EXAMPLE OF A QUARTERLY REVIEW AGENDA

Opening Comments	Site Manager
Project A Review	
Milestone Status	Project A Manager
Financial Status	Finance Manager
Issues	Project A Manager
Staff Comments	Finance and SQA
Project B Review	
Milestone Status	Project B Manager
Financial Status	Finance Manager
Issues	Project B Manager
Staff Comments	Finance and SQA
Project C Review	
Milestone Status	Project C Manager
Financial Status	Finance Manager
Issues	Project C Manager
Staff Comments	Finance and SQA
Remaining Project Overviews	
Milestone Status	Development Manager
Financial Status	Finance Manager
Issues	Development Manager
Staff Comments	Finance and SQA
Computing Support	
Performance Measures	Computing Manager
Issues	Computing Manager
Project Comments	Development Manager
Process Status	
Assessment Update	SEPG Manager
Action Plan Status	SEPG Manager
Technology Plan Status	SEPG Manager
Organization Performance	
Productivity	Finance/SEPG
Quality	SQA/SEPG
Action Item Summary	Site Manager

The Quarterly Review

- Figure 5.1 Typical SW Laboratory Organization

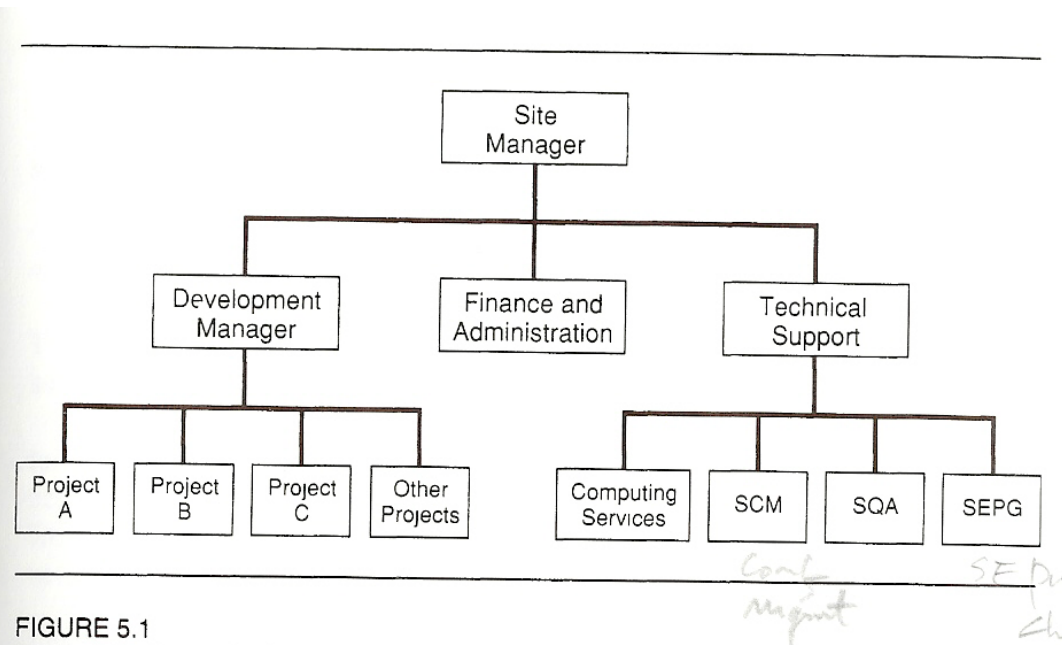
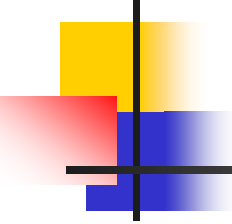


FIGURE 5.1
Typical Software Laboratory Organization

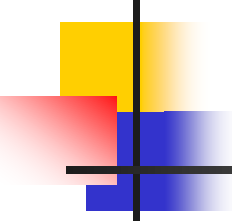


The Quarterly Review

- Table 5.2 Simple Project Schedule Checkpoints

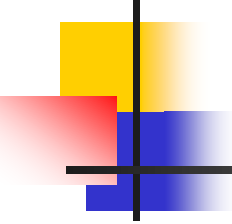
TABLE 5.2
SIMPLE PROJECT SCHEDULE CHECKPOINTS

Project: _____	Manager: _____	Date: _____		
Item	Planned	Projected	Actual	Difference
Functional Rqts. Approval	_____	_____	_____	_____
Phase 1 Review Approval	_____	_____	_____	_____
Design Reviews Complete	_____	_____	_____	_____
Phase 2 Review Approval	_____	_____	_____	_____
Code Reviews Complete	_____	_____	_____	_____
Unit Test Complete	_____	_____	_____	_____
Phase 3 Review Approval	_____	_____	_____	_____
Integration Test Entry Complete	_____	_____	_____	_____
System Test Complete	_____	_____	_____	_____
Phase 4 Review Approval	_____	_____	_____	_____
Customer Shipment	_____	_____	_____	_____



Project Phase Reviews

- Management must also **assess project progress periodically**.
- This is typically accomplished **through a sequence of reviews at key points** in each project.

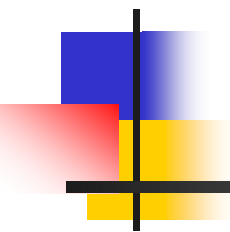


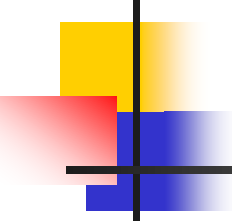
Project Phase Reviews

■ ISO/DoD Standards

- DoD standards MIL-STD-2167
 - Establishes a uniform SW development process which is applicable throughout the system life cycle
 - MIL-STD-498: SW Development and Documentation
- ISO/IEC 12207: Information Technology – SW Life Cycle Processes
- ISO/IEC 15504: Information Technology – Process Assessment (SPICE)
- ISO/IEC 15288: Systems and Software Engineering – System Life Cycle Processes
- ISO 9000 Series – standards related to quality management systems

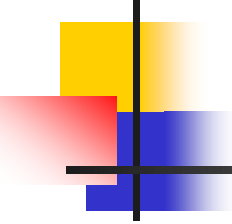
Establishing a Project Management System





Establishing a Project Management System

- The most important **management review system** is that conducted by **the project management team itself**.
- If project management is not aware of and actively involved in project issues, no other management system can be fully effective.
- With effective and involved project managers, however, a **management system** is still needed to **resolve conflicts between the projects and between the line and the staff organizations**.



References

- Humphrey, Watts S., *Managing the Software Process*, The SEI Series in Software Engineering, Addison-Wesley, 1989. (29th Printing, May 2003) (ISBN 0-201-18095-2)